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Simulation of nonlinear interactions between waves and floating bodies through a finite-element-based numerical tank

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Fully nonlinear water-wave interactions with a floating structure are investigated through a numerical towing tank. A wave maker is installed on one end of the tank while a numerical beach based on a combination of damping zone and Sommerfeld condition is adopted on the other side of the tank. A floating body is placed at a location in the tank, where it will be set into motion by the waves generated by the wave maker. The simulation is based on the velocity potential theory together with the finite-element method. The mesh used follows the deformation of the free surface and the body motion. Its structure is adjusted and the distribution of elements is completely rearranged when the motion is big to avoid an over-distorted grid. Auxiliary functions are introduced to decouple the nonlinear mutual dependence between the hydrodynamic force and the body motion. Extensive numerical results are provided for vertical circular cylinders and a simplified floating production, storage and offloading, for which meshes are obtained through an efficient scheme based on a two-dimensional tri-tree method.

Keywords: nonlinear wave/body interactions; numerical tank; finite-element method; tri-tree based hybrid mesh; bottom-mounted and truncated cylinders; FPSO

1. Introduction

It is often that an offshore structure or a ship is designed to operate in a very hostile environment. Both the free surface and the body motions can be quite large in this case. The traditional linear theory and the second-order theory used to predict these motions become invalid and the fully nonlinear method should be adopted. The essence of the method is that the computational domain will follow the deformation of the free surface and the motion of the body, and conditions are satisfied on the exact location of the boundary.

Water waves are very much dominated by gravity. In many cases, the viscosity will have significant effects only after many wave periods or many wavelengths. Even when there is a body in its path, the viscosity may have significant effects on the flow near the body, but its overall effects on the body motion and wave propagation are still small, if the body is 'large' based on the comparison of its typical dimension with the typical length and height of the wave. Thus it is very often the case that the velocity potential flow theory is used to analyse wave/body interactions.

The fully nonlinear two-dimensional problem was first considered by Longuet-Higgins & Cokelet (1976) and followed by Faltinsen (1977), Vinje & Brevig (1981) and Lin *et al*. (1984) using the boundary-element method (BEM). Applications in two dimensions based on the finite-element method (FEM) were performed by Wu & Eatock Taylor (1994, 1995), Clauss & Steinhagen (1999) and Robertson & Sherwin (1999). Recently, a coupled method was developed by Wu & Eatock Taylor (2003).

The three-dimensional applications based on the BEM include those by Ferrant (1994) and Celebi *et al*. (1998). It was argued, however, by Wu & Eatock Taylor (1995), through their own work on both BEM and FEM, that the latter is usually more computationally efficient, because of its much lower memory requirement. Thus extensive simulations have been undertaken using FEM for a variety of problems. These include sloshing in a three-dimensional tank (Wu *et al*. 1998), wave interaction with fixed cylinders (Ma *et al*. 2001*a*, *b*) and wave radiation by a cylinder in the forced motion (Hu *et al*. 2002*a*). Extensive comparison with experimental data is made, including those from Nestegard (1999), Clauss & Steinhagen (1999) and Retzler *et al*. (2000). Very good agreement is found.

These works have advanced our understanding of the nature of nonlinear interactions between waves and a body, but there are two major limitations in these applications. The first one is that the body is either fixed or in a forced motion. The second limitation is that the mesh structure or the connectivity between elements remains unchanged during the simulation, although the shape of each element changes due to the wave and body motions. The difficulty associated with the first limitation is due to the nonlinear coupling between the body motion and the force, i.e. the force depends on the acceleration and the acceleration, in turn, depends on the force. The difficulties associated with the second limitation is that, when a completely new mesh is generated, the information stored on the old mesh has to be transported smoothly to the new one. In the present work these two limitations have been successfully removed, which allows us to undertake simulation for more practical and relevant problems.

The first limitation is removed by using the scheme developed by Wu & Eatock Taylor (1996, 2003), who introduced some auxiliary functions to decouple the mutual dependence between the force and the body motion. The scheme has been implemented for the two-dimensional problem (Wu & Eatock Taylor 2003) and has been found to be very successful. A scheme is also developed to remove the second limitation. When a new mesh is generated, a search is conducted to locate each of its nodes in the old mesh. Information stored on the old mesh is then transported to the nodes of the new mesh through an interpolation. The test cases performed show both schemes are successful. Results have been provided for vertical cylinders and a simplified floating production, storage and offloading (FPSO).

2. A hybrid mesh generation

An essential part of computational fluid dynamics (CFD) simulation is the mesh generation. It is particularly important for the present case because of the deformation of the computational domain. For a realistic simulation, several thousand time-steps are usually required. It means that the mesh generation and the solution procedure should not take more than a few minutes at each step, if prohibitive CPU requirement is to be avoided. For this reason, the simulation in this work focuses

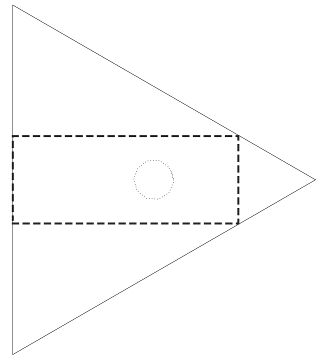

Figure 1. Seeding points.

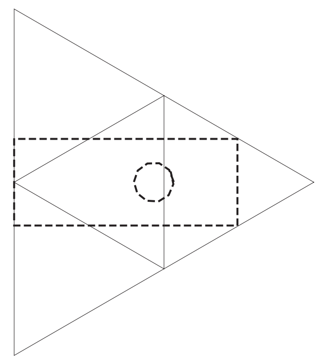

Figure 2. First division.

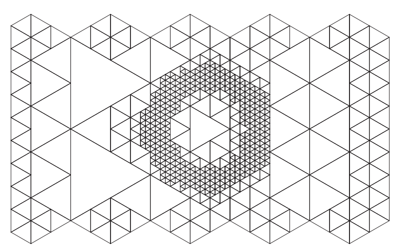

Figure 3. Seventh division.

on cylindrical structures, although the CFD code can deal with an arbitrary shape. Because the cross-section does not change in the vertical direction, it is possible to generate a two-dimensional mesh for the cross-section and then draw lines along the vertical directions to form the three-dimensional mesh. It should be emphasized, however, that there is no restriction on the shape of the cross-section of the body.

The two-dimensional mesh is generated through a tri-tree-based technique. The details can be found in Hu *et al*. (2002*b*). Here, we give brief a summary.

(1) Define an initial equilateral triangle, within which the desired fluid domain will lie and define a set of seeding points about which the mesh will be generated (figure 1).

(2) If the triangle contains a seeding point, divide the triangle. Otherwise, move

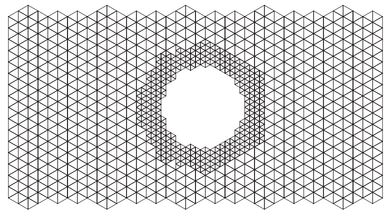

Figure 4. Application of minimum level 5.

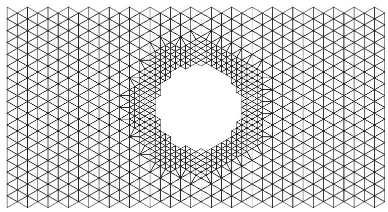

Figure 5. Mesh after face regulation and elimination of hanging nodes.

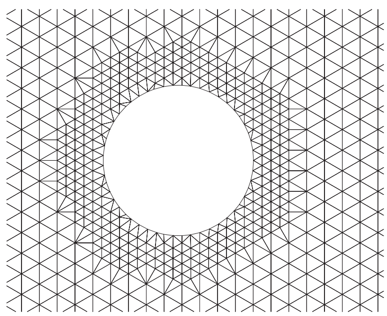

Figure 6. The detail of mesh after boundary treatment and corner regulation.

to next one, and repeat until the maximum division level has been reached (figures 2 and 3).

(3) Subdivide all elements to a minimum level to avoid very big elements due to lack of seeding points in the region (figure 4).

(4) Apply face regulation to restrict the ratio of triangle sides sharing a common edge to 2:1 and eliminate hanging nodes which reside at the middle of the edge of an adjacent element (figure 5).

(5) Apply boundary treatment around interior boundaries by moving nearby nodes to the surface of the boundary (figure 6).

After above procedures and some other refinements discussed in Hu *et al*. (2002*b*), the completed tri-tree mesh is shown in figure 7. The numbering system of the mesh is set up by giving each element an integer obtained through a mathematical formula. From the integer, the neighbours of the elements, its parents and children can be found.

The method is efficient when the dimensions in two directions of the computational domain are comparable. For a long and thin domain, such as the wave tank, the scheme becomes inefficient because much of the work is carried out in an area

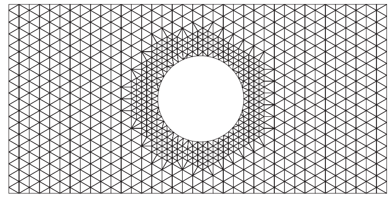

Figure 7. Unstructured tri-tree mesh.

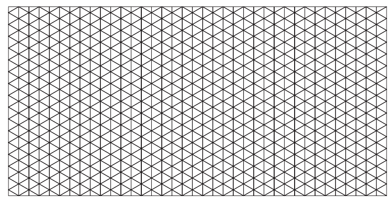

Figure 8. Structured mesh away from the body.

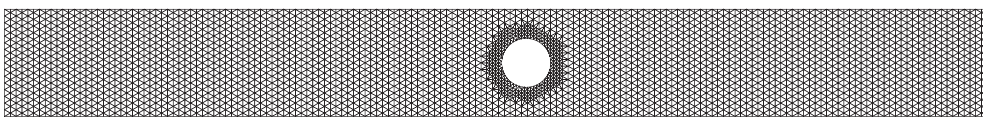

Figure 9. Hybrid mesh on the free surface around a cylinder.

with no real interest. On the other hand, we need a special mesh generator only in a region near the body where the domain can be complicated. Away from the body, the domain is usually regular and the grid can be obtained easily by a simple mesh generator. Thus we divide the whole domain into a near region and an outer region, and apply different mesh generators in these regions. The two meshes are then combined and their indexes are unified. Figure 8 shows the structured mesh away from the body. The combined hybrid meshes for a section of a single cylinder and for a section of an FPSO are given in figures 9 and 10.

After the two-dimensional mesh is generated, the three-dimensional mesh is obtained through the following steps.

(1) Extend the two-dimensional tri-tree plane in the third direction forming the three-dimensional mesh with prismatic elements.

(2) Split each individual prismatic element into three tetrahedral elements.

Once the three-dimensional mesh is generated, its index system is simultaneously established.

It is obvious that the above procedure for the three-dimensional mesh is only for a bottom-mounted body, as the vertical lines are drawn from the water surface to the bottom of the tank. For a truncated body, additional effort is needed. We note that the two-dimensional tri-tree mesh generator gives mesh both in the fluid domain and inside the body initially. But the internal elements are then eliminated. For a truncated cylinder, these 'internal' elements below the bottom of cylinder must be retained. Thus the region from the water surface to the bottom of the body and the region from the bottom of the body to the bottom of the tank have to be treated differently in the three-dimensional mesh generation. The index system has to be

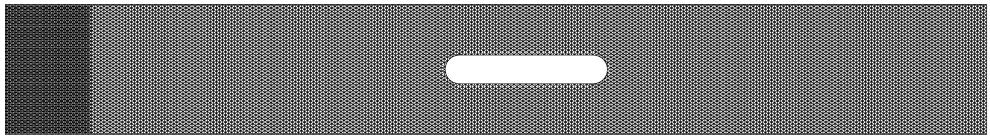

Figure 10. Hybrid mesh on the free surface around FPSO.

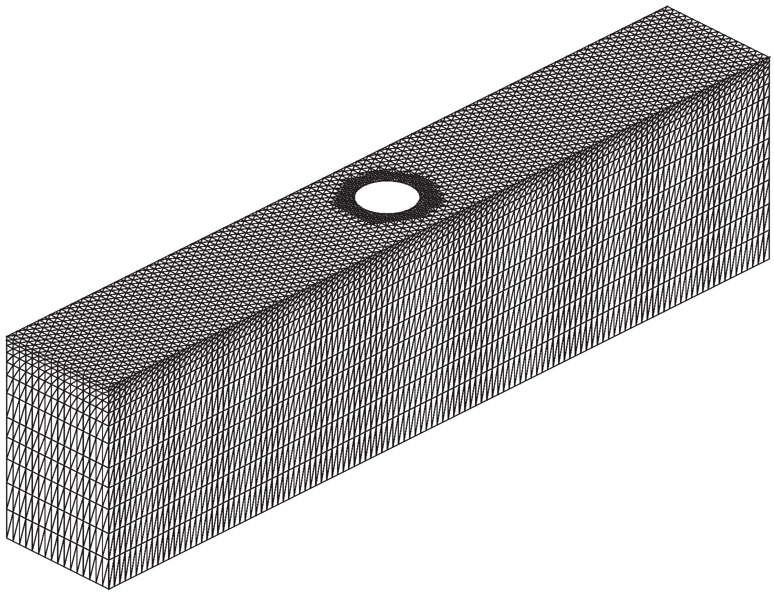

Figure 11. Three-dimensional mesh for one cylinder.

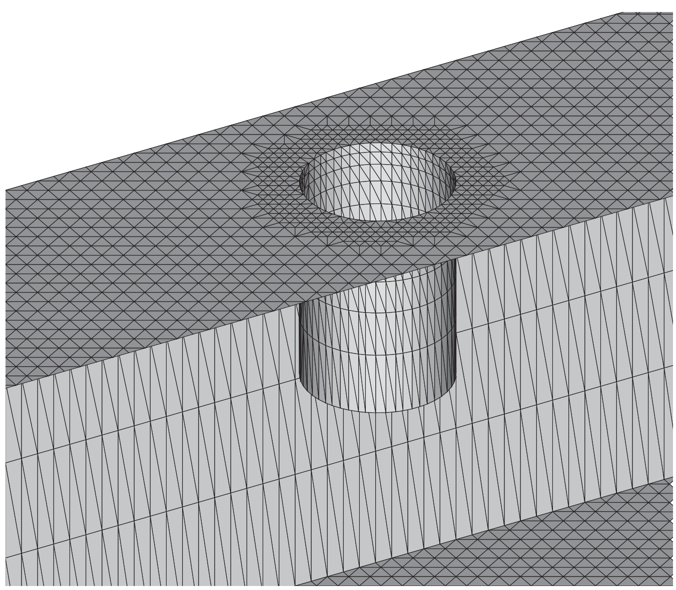

Figure 12. Surface mesh for a truncated cylinder.

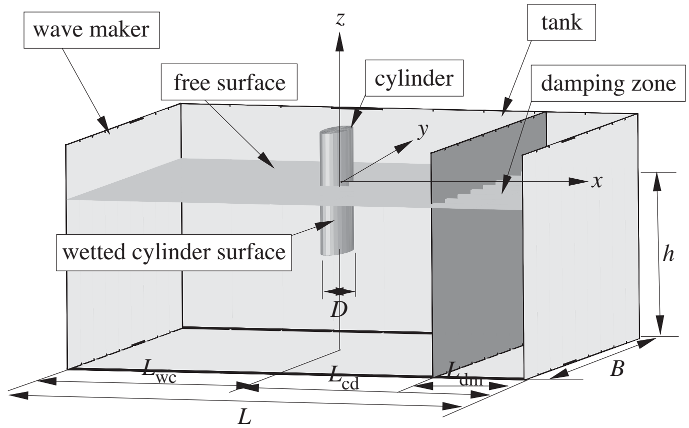

Figure 13. Computational domain for a numerical tank.

adjusted accordingly. The completed tetrahedral mesh for one cylinder is shown in figure 11. The surface mesh for a truncated cylinder is shown in figure 12.

If the nodes are followed throughout the calculation, the configuration of elements will change during the time-marching process. As a result, elements may be distorted and the quality of the mesh will deteriorate. Thus remeshing is applied based on the above procedure after every several time-steps.

3. Mathematical model and numerical method

The computational domain is shown in figure 13. A Cartesian system Oxyz is defined with z pointing upwards and the origin being on the mean free surface. The simulation is based on the velocity potential flow theory. The potential ϕ satisfies Laplace's equation:

$$\nabla^2\phi = 0. \tag{3.1}$$

In terms of Lagrangian description, the kinematic and dynamic conditions on the free surface can be written as

$$\frac{\mathrm{D}x}{\mathrm{D}t} = \frac{\partial\phi}{\partial x}, \qquad \frac{\mathrm{D}y}{\mathrm{D}t} = \frac{\partial\phi}{\partial y}, \qquad \frac{\mathrm{D}z}{\mathrm{D}t} = \frac{\partial\phi}{\partial z}, \tag{3.2}$$

$$\frac{\mathrm{D}\phi}{\mathrm{D}t} = -gz + \tfrac{1}{2}\nabla\phi\nabla\phi, \tag{3.3}$$

where g is the acceleration due to the gravity. The condition on the wave maker can be expressed as

$$\frac{\partial\phi}{\partial n} = \boldsymbol{U}_0 \cdot \boldsymbol{n}, \tag{3.4}$$

where $\boldsymbol{U}_0$ is the velocity of the wave maker and $\boldsymbol{n}$ is its normal vector pointing out of the fluid domain. The boundary condition on the moving body surface is

$$\frac{\partial\phi}{\partial n} = \boldsymbol{V} \cdot \boldsymbol{n}, \tag{3.5}$$

where $\boldsymbol{V}$ is the velocity of the body. On the fixed boundary, such as side walls and the bottom of the tank, the condition is

$$\frac{\partial \phi}{\partial n} = 0. \tag{3.6}$$

The radiation condition at the far end is imposed through a damping zone and Sommerfeld condition (Ma *et al*. 2001*a*).

The simulation starts at an instant, say, $t = 0$, at which it is assumed that the shape of the free surface and the potential on the free surface are known. The boundary-value problem is then complete and can be solved. Equations (3.2) and (3.3) can be used to update the information on the free surface, the solution procedure moves to the next step.

At each time-step, the problem may be solved by the FEM. The velocity potential ϕ can be written in terms of a shape function, $N_J(x, y, z)$,

$$\phi = \sum_J \phi_J N_J(x, y, z), \tag{3.7}$$

where ϕ_J is the velocity potential at node J. Using the Galerkin method, we have

$$\iiint_\forall \nabla^2 \phi N_i \, \mathrm{d}V = 0. \tag{3.8}$$

Using Green's identity and the boundary conditions, this equation becomes

$$\begin{aligned} &\iiint_\forall \nabla N_I \cdot \sum_{\substack{J \\ J \notin S_\mathrm{p}}} \phi_J \nabla N_J \, \mathrm{d}V \\ &\qquad = \iint_{S_\mathrm{n}} N_I f_\mathrm{n} \, \mathrm{d}S - \iiint_\forall \nabla N_I \cdot \sum_{\substack{J \\ J \in S_\mathrm{p}}} (f_\mathrm{p})_J \nabla N_J \, \mathrm{d}V, \quad I \notin S_\mathrm{p}, \end{aligned} \tag{3.9}$$

where S_p represents the Dirichlet boundary on which the velocity potential, denoted by f_p, is known, and S_n represents the Neumann boundary on which the normal derivative of the velocity potential, denoted by f_n, is known. The equation can further be written in the matrix form

$$[A]\{\phi\} = [B], \tag{3.10}$$

where

$$\begin{aligned} \{\phi\} &= [\phi_1, \phi_2, \phi_3, \ldots, \phi_I, \ldots]', && I \notin S_\mathrm{p}, \\ A_{IJ} &= \iiint_\forall \nabla N_I \cdot \nabla N_J \, \mathrm{d}\forall, && I \notin S_\mathrm{p} \text{ and } J \notin S_\mathrm{p}, \\ B_I &= \iint_{S_\mathrm{n}} N_I f_\mathrm{n} \, \mathrm{d}S - \iiint_\forall \nabla N_I \sum_{\substack{J \\ J \in S_\mathrm{p}}} (f_\mathrm{p})_J \nabla N_J \, \mathrm{d}\forall, && I \notin S_\mathrm{p}. \end{aligned}$$

This linear matrix equation is then solved through iteration with a symmetric successive over relaxation (SSOR) preconditioner. Once the potential is found, the velocity is obtained by taking derivatives of the potential along the lines linking the current node with the neighbouring nodes, together with the least-square method. The recovery technique (Zienkiewicz & Zhu 1992) is then used to improve the accuracy of the result.

The body motion is obtained through Newton's law. The force $\boldsymbol{F}$ and the moment $\boldsymbol{M}$ acting on the body can be found by integrating the pressure over the wetted surface,

$$\boldsymbol{F} = -\rho \iint_{S_\mathrm{b}} (\phi_\mathrm{t} + \tfrac{1}{2}\nabla\phi\nabla\phi + gz)\boldsymbol{n}\,\mathrm{d}S, \tag{3.11}$$

$$\boldsymbol{M} = -\rho \iint_{S_\mathrm{b}} (\phi_\mathrm{t} + \tfrac{1}{2}\nabla\phi\nabla\phi + gz)(\boldsymbol{X}\times\boldsymbol{n})\,\mathrm{d}S, \tag{3.12}$$

where ρ is the density of the fluid, $\boldsymbol{X}$ is the position vector starting from the centre of rotation and S_b is the body surface. If we decompose the body motion into translation $\boldsymbol{U}$ and rotation $\boldsymbol{\Omega}$, the equation for the motion can be written in the matrix form

$$[M_\mathrm{b}][\dot{U}] = [F] + [F_\mathrm{e}], \tag{3.13}$$

where $[M_\mathrm{b}]$ is the body mass matrix, $[\dot{U}]$ is a column of body acceleration with three translations and three rotations, $[F]$ is a column of hydrodynamic force and moment from (3.11), and $[F_\mathrm{e}]$ is a column of the external force such as the force due to the gravity. It may appear that equation (3.13) is straightforward, but the difficulty is with ϕ_t, which is not explicitly given even when ϕ itself is known. It might be possible to find ϕ_t explicitly from the backward difference method, but this is not always accurate enough and it may sometimes lead to numerical stability in the calculation of the body motion. Thus we use the scheme developed by Wu & Eatock Taylor (1996, 2003), which is summarized below in a slightly different format.

In the fluid domain, the time derivative ϕ_t satisfies the Laplace equation

$$\nabla^2\phi_\mathrm{t} = 0. \tag{3.14}$$

On the fixed boundary, it satisfies

$$\frac{\partial\phi_\mathrm{t}}{\partial n} = 0. \tag{3.15}$$

On the free surface, ϕ_t is given by the Bernoulli equation as

$$\phi_\mathrm{t} = -gz - \tfrac{1}{2}\nabla\phi\nabla\phi. \tag{3.16}$$

On the moving boundary of the body surface, the condition can be written as (Wu 1998)

$$\frac{\partial}{\partial n}(\phi_\mathrm{t}) = [\dot{\boldsymbol{U}} + \dot{\boldsymbol{\Omega}}\times\boldsymbol{X}]\cdot\boldsymbol{n} - \boldsymbol{U}\cdot\frac{\partial\nabla\phi}{\partial n} + \boldsymbol{\Omega}\cdot\frac{\partial}{\partial n}[\boldsymbol{X}\times(\boldsymbol{U}-\nabla\phi)] \tag{3.17}$$

and on the wave maker the condition becomes

$$\frac{\partial}{\partial n}(\phi_\mathrm{t}) = \dot{\boldsymbol{U}}_0\cdot\boldsymbol{n} - \boldsymbol{U}_0\cdot\frac{\partial\nabla\phi}{\partial n}. \tag{3.18}$$

We consider only the translation here because of the restriction of the mesh generator. Thus $\boldsymbol{\Omega} = \boldsymbol{0}$ and $\boldsymbol{V} = \boldsymbol{U}$. Equation (3.17) becomes

$$\frac{\partial}{\partial n}(\phi_\mathrm{t}) = \dot{\boldsymbol{U}}\cdot\boldsymbol{n} - \boldsymbol{U}\cdot\frac{\partial\nabla\phi}{\partial n}. \tag{3.19}$$

If the acceleration of the body $\dot{\boldsymbol{U}}$ in equation (3.19) were known, ϕ_{t} could be solved in a manner similar to that used for the velocity potential ϕ. The difficulty, however, is that the acceleration is not known before ϕ_{t} is found. We then divide ϕ_{t} into two parts,

$$\phi_{\mathrm{t}} = Q + G.$$

On the body surface, equation (3.19) can be split as

$$\frac{\partial G}{\partial n} = -\boldsymbol{U} \cdot \frac{\partial \nabla \phi}{\partial n}, \tag{3.20}$$

$$\frac{\partial Q}{\partial n} = \dot{\boldsymbol{U}} \cdot \boldsymbol{n}. \tag{3.21}$$

On the free surface, ϕ_{t} in (3.16) gives

$$G = -gz - \tfrac{1}{2}\nabla\phi\nabla\phi, \tag{3.22}$$

$$Q = 0. \tag{3.23}$$

On the wave maker, equation (3.18) becomes

$$\frac{\partial G}{\partial n} = \dot{\boldsymbol{U}}_0 \cdot \boldsymbol{n} - \boldsymbol{U}_0 \cdot \frac{\partial \nabla \phi}{\partial n}, \tag{3.24}$$

$$\frac{\partial Q}{\partial n} = 0. \tag{3.25}$$

The force acting on the body in (3.11) can be rewritten as

$$F_j = -\rho \iint_{S_{\mathrm{b}}} (G + \tfrac{1}{2}\nabla\phi\nabla\phi + gz) n_j \,\mathrm{d}S - \rho \iint_{S_{\mathrm{b}}} Q \cdot n_j \,\mathrm{d}S. \tag{3.26}$$

We further define $G = \gamma - \boldsymbol{U}\nabla\phi$. Substituting G into equation (3.20) yields

$$\frac{\partial \gamma}{\partial n} = 0 \tag{3.27}$$

on the body surface. Substituting expression for G into equation (3.22) yields

$$\gamma = \boldsymbol{U}\nabla\phi - gz - \tfrac{1}{2}|\nabla\phi|^2 \tag{3.28}$$

on the free surface. On other surfaces such as the wave maker and bottom, the boundary condition can be obtained from

$$\frac{\partial \gamma}{\partial n} = \boldsymbol{U} \cdot \frac{\partial \nabla \phi}{\partial n}. \tag{3.29}$$

The transformation of G to γ is a transform of the double derivatives in the condition on the body surface to those on the side wall, the bottom and the wave maker, where the results are either zero or can be obtained more easily. All the boundary conditions on γ are known once the potential is found and it can be solved in a manner similar to that used for ϕ. In equation (3.26), however, Q is still unknown, as the acceleration of the body required in (3.21) is not given in advance and it depends on the hydrodynamic force $\boldsymbol{F}$.

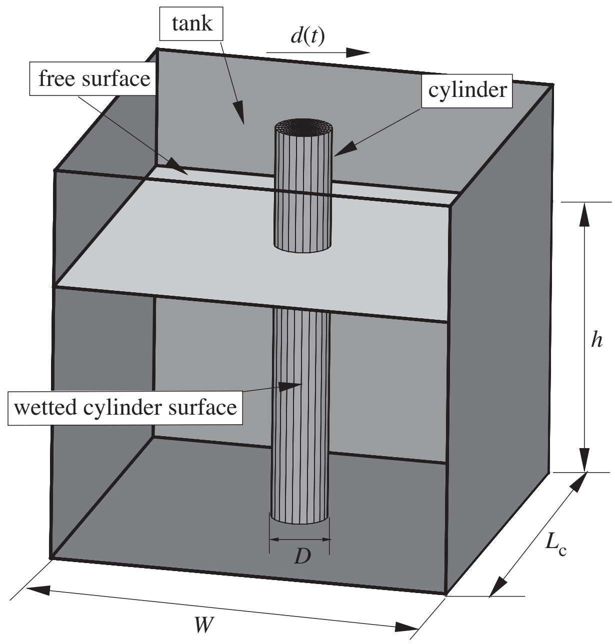

Figure 14. Computational domain for figures 15 and 16.

We introduce some auxiliary functions χ_i, defined as $Q = \dot{U}_1\chi_1 + \dot{U}_2\chi_2 + \dot{U}_3\chi_3$ (Wu & Eatock Taylor 2003). We require these functions to satisfy the Laplace equation in the fluid domain,

$$\nabla^2\chi_i = 0, \tag{3.30}$$

the boundary condition

$$\frac{\partial\chi_i}{\partial n} = n_i \tag{3.31}$$

on the body surface and

$$\chi_i = 0 \tag{3.32}$$

on the free surface. The condition on the wave maker and other boundaries can be expressed as

$$\frac{\partial\chi_i}{\partial n} = 0. \tag{3.33}$$

The second term of the right-hand side in equation (3.26) can be expressed as

$$\begin{aligned} &-\rho\iint_{S_\mathrm{b}} Q\cdot n_j\,\mathrm{d}S \\ &\qquad = -\rho\dot{U}_1\iint_{S_\mathrm{b}}\chi_1\cdot n_j\,\mathrm{d}S - \rho\dot{U}_2\iint_{S_\mathrm{b}}\chi_2\cdot n_j\,\mathrm{d}S - \rho\dot{U}_3\iint_{S_\mathrm{b}}\chi_3\cdot n_j\,\mathrm{d}S. \end{aligned} \tag{3.34}$$

Substituting this equation into (3.26) and combining this with (3.13), we have

$$\sum_{i=1}^{3}(\delta_{ji}M_\mathrm{b} + N_{ji})\cdot\dot{U}_i = f_j - M_\mathrm{b}g\delta_{j3}, \tag{3.35}$$

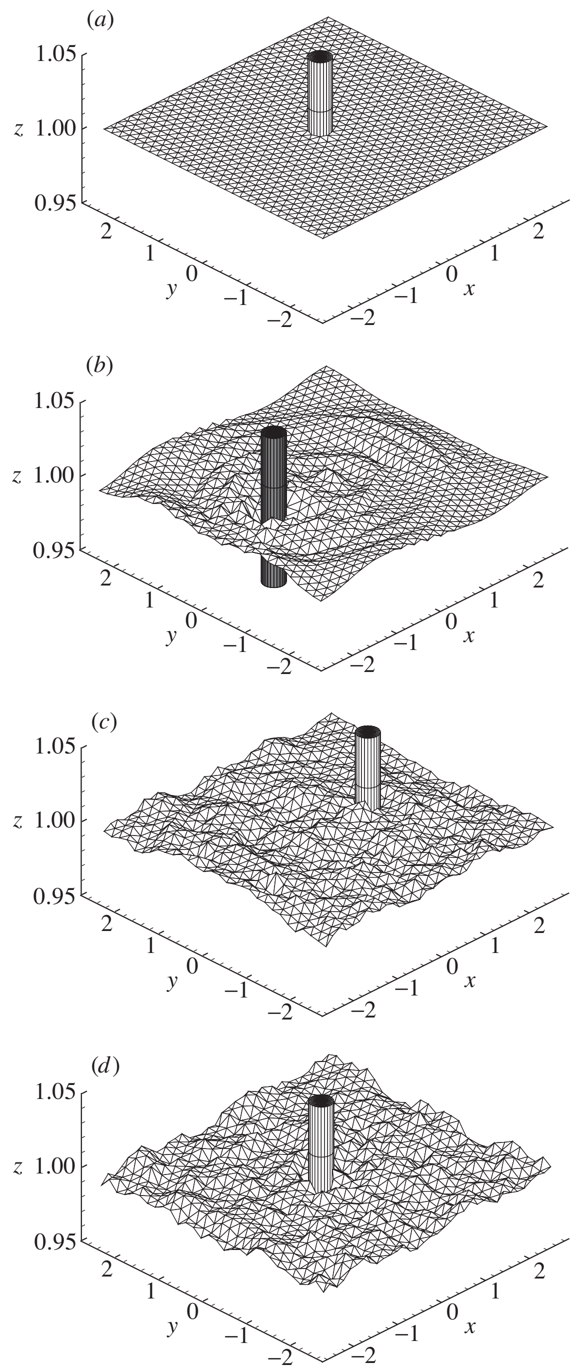

Figure 15. Wave profile by a cylinder undergoing periodic oscillation: (a) $t = 0$; (b) $t = T/4$; (c) $t = 3T/4$; (d) $t = T$.

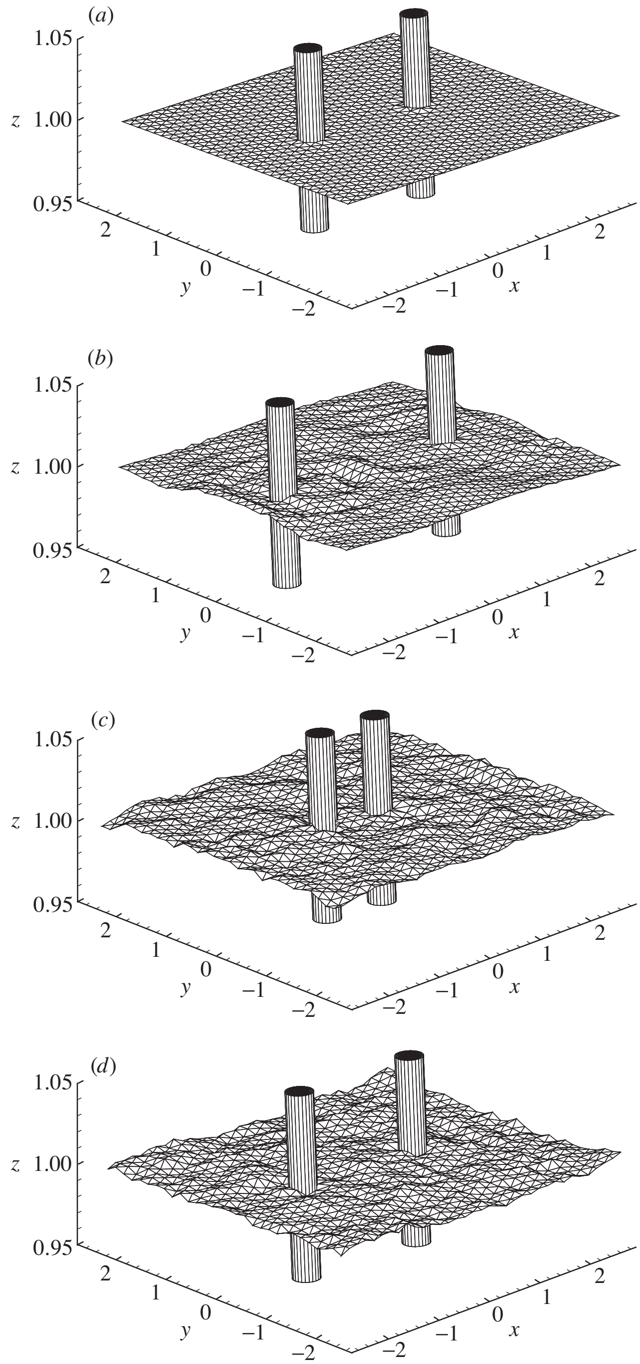

Figure 16. Wave profile by two cylinders undergoing periodic oscillation: (*a*) $t = 0$; (*b*) $t = T/4$; (*c*) $t = 3T/4$; (*d*) $t = T$.

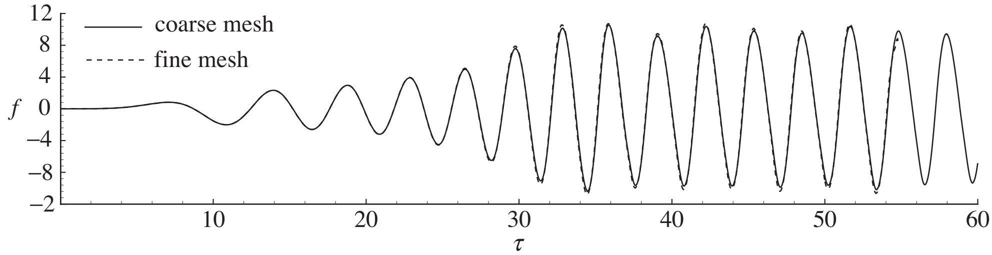

Figure 17. Forces from coarse and fine meshes at $a = 0.02$.

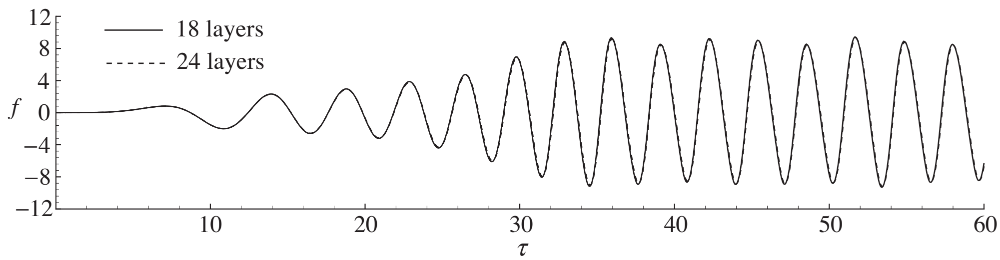

Figure 18. Forces from different number of layers of elements in vertical direction at $a = 0.02$.

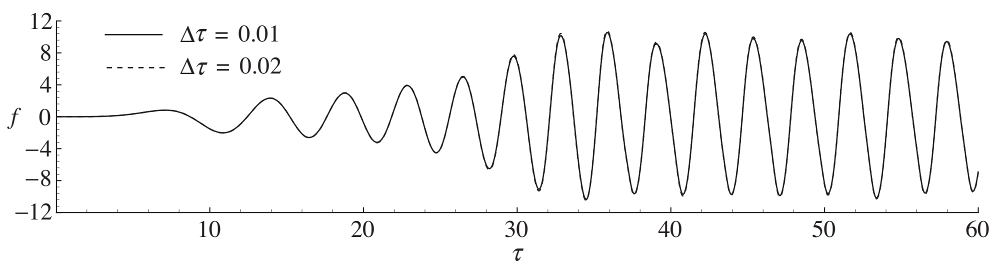

Figure 19. Forces from different time-steps at $a = 0.02$.

where

$$N_{ji} = \rho \iint_{S_\mathrm{b}} \chi_j \cdot n_i \,\mathrm{d}S, \qquad f_j = -\rho \iint_{S_\mathrm{b}} (G + \tfrac{1}{2}\nabla\phi\nabla\phi + gz) n_j \,\mathrm{d}S$$

and $\delta_{ji} = 1$ $(j = i)$ and $\delta_{ji} = 0$ $(j \neq i)$.

Equation (3.35) shows that the components of the body acceleration $\dot{\boldsymbol{U}}$ in the x-, y- and z-directions can be obtained directly before the force on the body is known. After the acceleration has been found, $\boldsymbol{F}$ can be found from equation (3.26). The velocity and the position of the body can also be updated.

The computational procedure can be summarized as follows.

(a) Generate the three-dimensional tetrahedral mesh near the body based on the tri-tree method.

(b) Generate the three-dimensional structured mesh away from the body.

(c) Merge the two mesh systems and unify the indexes.

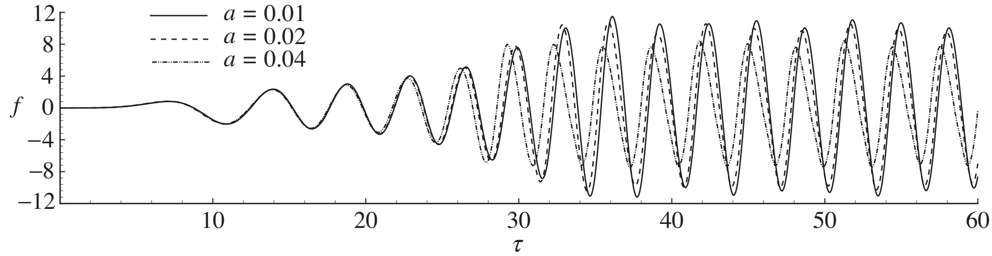

Figure 20. Force on the bottom-mounted cylinder.

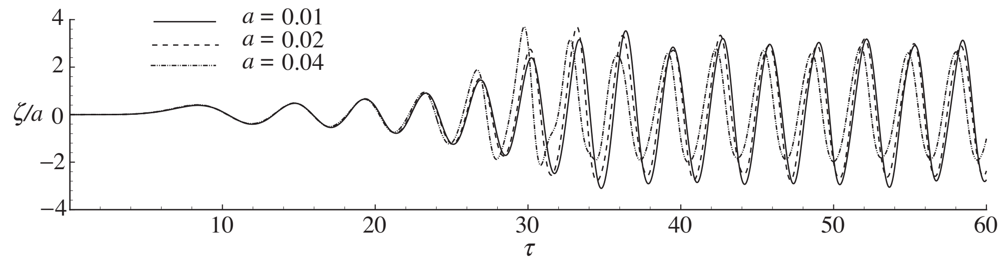

Figure 21. Time-history of wave run up on the front side of the bottom-mounted cylinder.

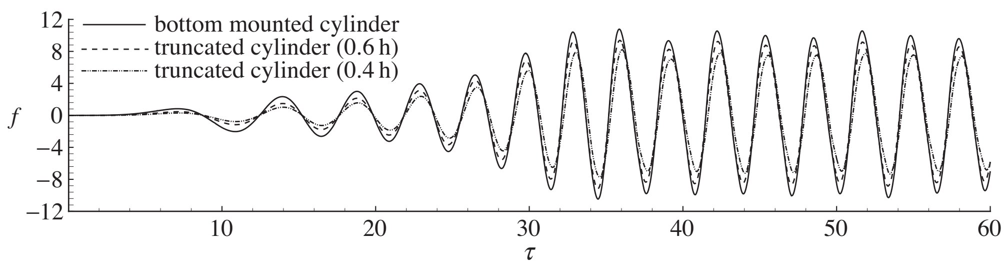

Figure 22. Force on cylinder with different draughts at $a = 0.02$.

(d) Calculate the global matrix $[A]$ and the column containing the boundary condition $[B]$.

(e) Solve the linear equation to obtain the velocity potentials ϕ_j and the auxiliary functions χ_j using the conjugate gradient iterative procedure with SSOR preconditioner.

(f) Calculate the fluid velocity on the free surface and the body surface.

(g) Use the recovery technique to improve the accuracy.

(h) Find γ and calculate the acceleration of the body with the help of the auxiliary functions χ_j.

(i) Find the new velocity and position of the body.

(j) Track the free surface through the movement of the fluid particle on the element nodes and update the values of the potential on these nodes.

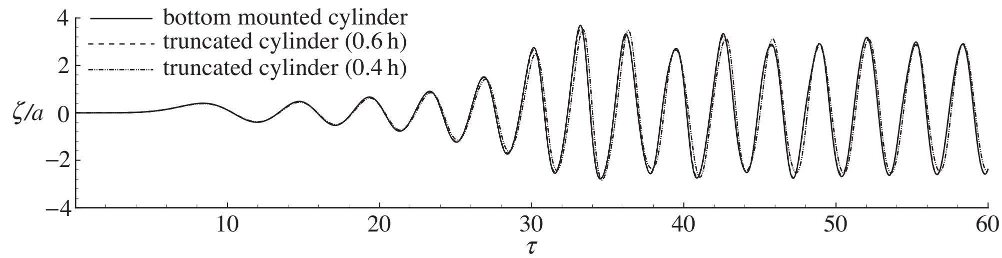

Figure 23. Time-history of wave run up on the front side of the cylinder with different draughts at $a = 0.02$.

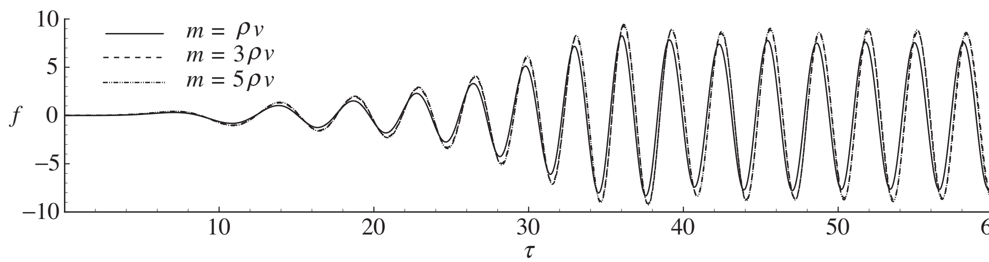

Figure 24. The time-history of the force on the floating cylinder at $a = 0.01$.

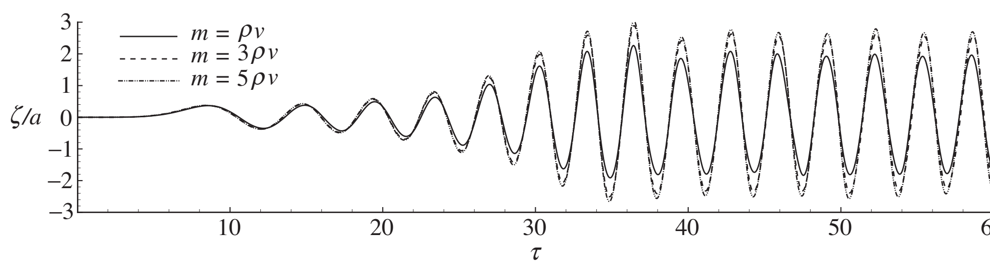

Figure 25. Time-history of wave run up on the front side of the floating cylinder at $a = 0.01$.

(k) Start the calculation for the next time-step. Go to (a) if remeshing is needed, and (d) if it is not.

4. Results

In all the simulations, the acceleration g due to gravity, the water depth h and the water density ρ are used for the non-dimensionalization. Calculation was first made for some of the cases in Ma *et al.* (2001*a*, *b*) and results were found in good agreement with those obtained there. We now consider an example where it would not be practical to fix the mesh structure and its connectivity. The fluid domain used here is a rectangular tank and is illustrated in figure 14. For the results given below, $L_c = W = 5.0$ and the diameter of the cylinder $D = 0.4$. The cylinder is in the forced motion with the displacement defined by $d(t) = -a\sin(\omega t)$, where a is the amplitude and taken as $a = 1.0$, ω is the excitation frequency and taken as $\omega = 0.1$. It is evident that the motion amplitude is sufficiently large that the elements on one side would be squeezed while those on the other side would be over stretched if the connectivity

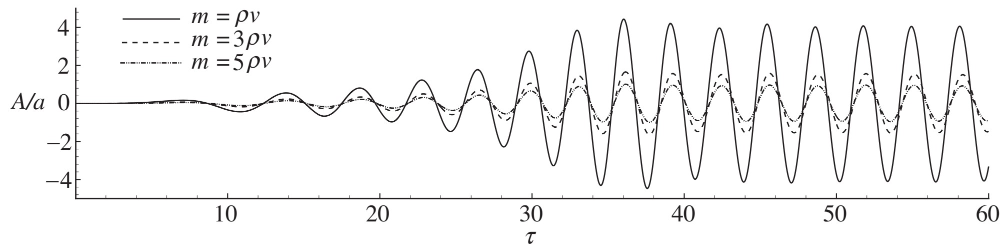

Figure 26. Time-history of the body acceleration at $a = 0.01$.

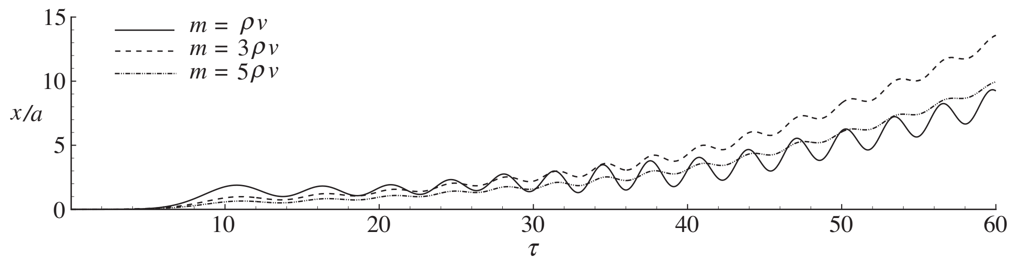

Figure 27. Time-history of the body motion at $a = 0.01$.

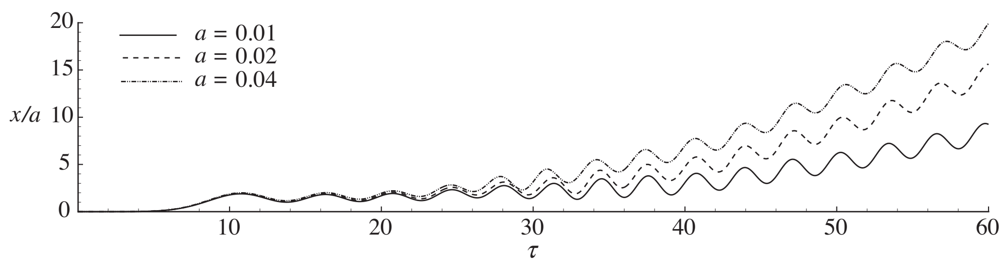

Figure 28. Time-history of the body motion at different a ($m = \rho v$).

of the mesh remains unchanged. In the present work, remeshing is applied regularly and the information is transported from mesh to mesh smoothly.

The initial number of tetrahedral elements used is *ca.* 71748 for a single cylinder. The CPU time needed for 3136 time-steps over one period is *ca.* 5 hours on a Compaq Workstation XP1000. Figure 15 presents the wave profiles over a full oscillation period. The wave elevation has been amplified by 27 times. It is clear from these profiles that a rational mesh structure has always been maintained when the cylinder moves towards or away for the wall with large amplitude. We calculated another case with two cylinders with $a = 0.5$ and $\omega = 0.1$, and the wave profiles are shown in figure 16 with the same amplification as that in figure 15. The initial distance between the centres of two cylinders is $5D$ (figure 16*a*), the longest distance between them is $7.5D$ (figure 16*b*), and the shortest distance is $2.5D$ (figure 16*c*). Together with figure 15, the result has demonstrated the capability of the scheme for large deformation of the domain.

We next consider the numerical tank illustrated in figure 13. In the following simulations for the cylinder, the initial position of the body is at a distance $L_{\mathrm{wc}} = 7.0$ from the wave maker. The distance between the cylinder and the far end is $L_{\mathrm{cd}} = 8.0$.

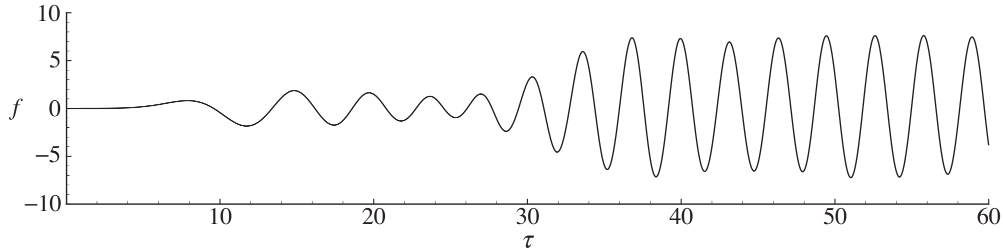

Figure 29. Time-history of the force on the FPSO at $a = 0.01$.

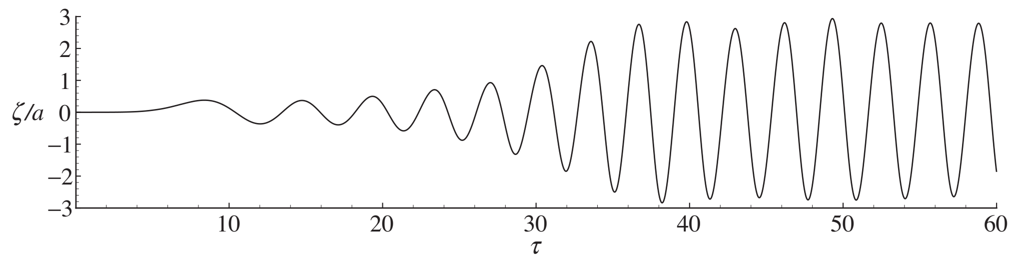

Figure 30. Time-history of wave run up on the front side of the FPSO at $a = 0.01$.

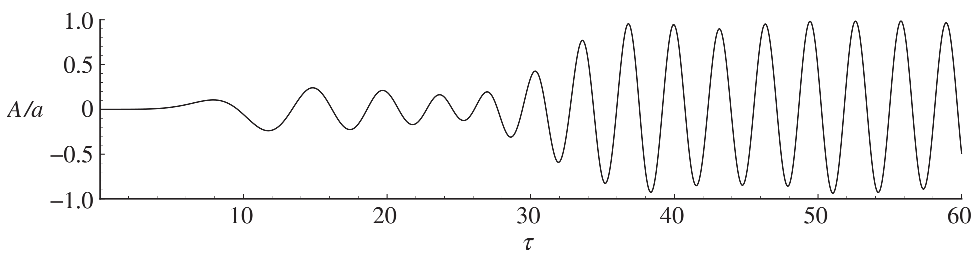

Figure 31. Time-history of the acceleration of the FPSO at $a = 0.01$.

The length of the damping zone is specified as $L_{\mathrm{dm}} = 3.0$. The diameter of the cylinder is taken as $D = 0.28$ and the tank width $B = 0.62$. The motion of the wave maker motion is defined as $d(t) = -a\cos(\omega t)$. The mesh is obtained through the procedure discussed in § 2. It should be noted that pattern of the mesh near the wave maker could have crucial implications to the transverse stability of the wave system. A suitable structure used by Wu *et al*. (1996) has been adopted here.

To check the accuracy of the numerical tank, coarser and finer meshes, with 17 804 elements and 22 835 elements on the horizontal plane, respectively, different numbers of layers of elements in the vertical direction and different time-steps are used. Figures 17–19 give the time-history of the force on a bottom-mounted cylinder with $a = 0.02$ and $\omega = 2$. The force in the figure is defined as $f = \mathrm{force}/R^2a$. It is evident that the result has converged with both the mesh and the time-step. Figure 20 gives the time-history of the force acting on the cylinder in the x direction, which is subjected to waves generated at $\omega = 2$ and three different amplitudes. The corresponding wave run up on the front side of the cylinder is shown in figure 21. The nonlinear effect in these figures is evident. Further simulation is made for two truncated cylinders with draught equal to $0.4h$ and $0.6h$, respectively. The force on

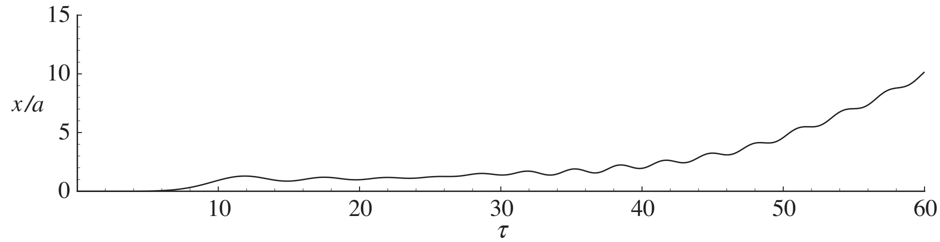

Figure 32. Time-history of the displacement of the FPSO at $a = 0.01$.

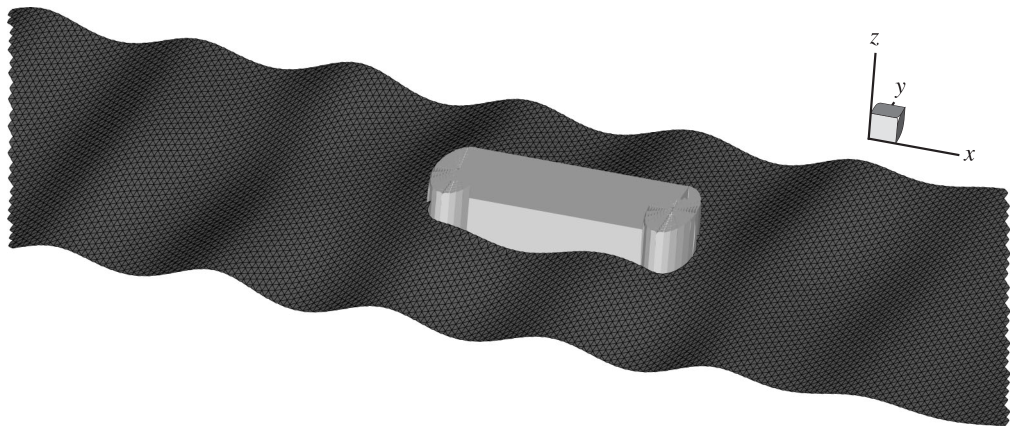

Figure 33. Wave profile around the FPSO.

these cylinders is given in figure 22 and the corresponding run up on the front side of the cylinder is shown in figure 23. It is noticeable that the draught has little effect on the wave elevation. It does have some effect on the force, but still not too significant if we notice that the cylinder has been cut by about one-half. This is, of course, not unexpected, as most of the wave force is near the water line.

Figures 24–27 give various results for a floating cylinder with draught equal to $0.6h$. The configuration of the tank is the same as that motioned before. The wave maker undergoes similar sinusoidal oscillation with $a = 0.01$ and $\omega = 2$. In these figures, ρv is the initial mass displacement of the cylinder and m is the body mass. The cylinder is allowed to have surge motion, responding to the wave excitation, while all other modes are restrained. As expected, the effect of the body weight on the force and run up is small. Its main effect is on the acceleration and the displacement, and the heavier the body, the smaller the acceleration. Figure 28 shows the motion history for a cylinder of weight $m = \rho v$ in waves corresponding to a different a. Apart from the periodic oscillation, the drift motion of the cylinder is evident. The motion is bigger when a is bigger, which is consistent with the physics of the problem.

Figures 29–32 give various results for a simplified FPSO, which is formed by a rectangle and two semicircles at two ends. The radius of the semicircle is taken as $R = 0.218$, the total length of the FPSO is taken as 2.464 and the draught is equal to $0.15h$. The weight of the ship is assumed to be equal to the initial weight displacement of the FPSO. The tank width $B = 2.0$ and the distance between the centre of the left semicircle of the FPSO and far end is $L_{\mathrm{cd}} = 10.0$. The rest of the configuration of the tank is the same as that for the cylinder. Figures 29–32 give the force, wave run up, acceleration and motion. The overall behaviour of these results is very similar to that for the cylinder. Figure 33 shows a wave profile around the FPSO. The result includes the incoming wave, diffracted wave, radiated wave and the wave created by the drift motion of the body.

5. Conclusion

A three-dimensional numerical tank has been developed based on the velocity potential flow theory using the FEM with the hybrid mesh generation. The computational domain follows the wave and body motion and fully nonlinear effects have been included. The full remeshing has been successfully implemented, which allows the mesh structure and its connectivity to evolve with time to retain the quality of the mesh. The method based on the auxiliary functions to decouple the mutual dependence between the force and the acceleration has been successfully implemented for the three-dimensional problems. It is evident, however, that the potential of the present work has been limited by the mesh generator, which clearly needs more work.

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